

Ricardo Kirkner

Staff Engineer | Backend Systems & Observability

Passionate about making complexity understandable

✉ ricardo@kirkner.com.ar 🌐 www.kirkner.com.ar 🐙 github.com/ricardokirkner 🔗 linkedin.com/in/ricardokirkner

Guiding Interests

Inspired by the awe that comes from understanding, I build infrastructure that **reveals structure in complexity**, supports **understanding and discovery**, and elevates how we reason, learn, and reflect. My motivation lies not only in scalability, but in creating tools that help humans and machines **make sense of the world in awe-inspiring ways**.

Professional Summary

Staff Engineer with 20+ years of experience architecting and building infrastructure platforms from the ground up. Specializes in greenfield system design, creating compelling proof-of-concepts, and leading technical implementation of complex distributed systems. Deep expertise in **observability platforms**, multi-cloud infrastructure, and modern backend architectures. Thrives on solving novel technical challenges and building systems that inspire awe.

Core Expertise: Greenfield System Architecture • Infrastructure Platform Design • Observability & Monitoring Systems • **OpenTelemetry & Standards** • Technical Leadership

Technical Skills

Core Languages: Go • Python

Infrastructure: Kubernetes • Docker • LXC/cgroups • GCP • AWS • Azure

Observability: OpenTelemetry • Grafana • Sentry • Nagios • Graphite

OS & Web: Linux (Debian, Fedora) • REST API Design • Django • Flask • Pyramid • HTML/CSS • Jinja2 • Mako

DevOps: GitLab CI/CD • Jenkins • Puppet • Ansible

Data & Storage: BigQuery • ClickHouse • PostgreSQL • MySQL • ETL Pipelines

Professional Experience

Platform.sh

2018–Present

Staff Engineer

Previously: Technical Lead (2019–2020), Cloud Engineer (2018–2019)

Platform Architecture & Implementation

- **Architected observability infrastructure from the ground up** using Go, designing comprehensive metrics collection, storage, and API systems for global platform serving **10,000+ projects**
- **OpenTelemetry Pioneer:** Early advocate and implementer of OpenTelemetry standards, driving adoption when recognizing limitations of existing proprietary solutions
- **Designed critical platform components** including edge proxy routing layer (HTTP/TCP/TLS) and orchestration control plane for multi-cloud infrastructure
- **Reduced observability COGS by 50-60%** through innovative architectural decisions and efficient system design
- **Led multi-region feature rollouts** across AWS, GCP, and Azure for **10,000+ projects**, architecting deployment strategies and ensuring system reliability

Infrastructure Innovation & Security

- **Designed and implemented SSH certificate authentication** and automated TLS certificate renewal systems
- **Architected logs forwarding infrastructure** and comprehensive observability feature set
- **Contributed to discovery of CVE-2024-45337** in golang x/crypto/ssh library, demonstrating deep system-level security expertise

Technical Leadership & Collaboration

- Mentor multiple engineers and created internal training materials on observability best practices and OpenTelemetry implementation
- Drive technical consistency and architectural standards across teams, with focus on modern observability practices
- Champion adoption of industry-standard monitoring approaches over proprietary solutions

Canonical

2010–2018

Software Engineer, Team Manager, Technical Lead

Ubuntu Platform Architecture

- **Designed Ubuntu SSO infrastructure** serving millions of users, designing scalable authentication systems
- **Built Ubuntu Pay payment processing** including in-app purchases for Snap packages, handling complex financial transactions
- **Led Software Center integration** with Snap package ecosystem, bridging traditional and modern package management
- Developed scalable authentication and payment processing systems using Python for Ubuntu ecosystem

Globant

2009–2010

Technical Lead

Massive-Scale Data Infrastructure

- **Architected data collection system** for Active Broadband Networks handling 17 million cable modem devices on Comcast network
- **Designed high-performance ETL pipelines** for processing massive data volumes while meeting strict performance and hardware constraints
- Led development team building custom solutions for large-scale analytics and delayed consumption processing

Early Career Highlights

2001–2009

Universidad de Buenos Aires

2001–2005

Network & System Administrator

- Managed faculty-wide network and system infrastructure serving needs for departments including Computer Science, Physics, and Mathematics

Various Roles

2005–2008

Software Engineer

- Developed Plone-based multi-lingual portal for Basque community worldwide
- Designed and lead development of Buenos Aires City Driver License Exam system serving 1500 applicants daily

Key Achievements

-
- **Architect of enterprise-scale platforms** - Built observability, authentication, and data processing systems from ground up serving 10,000+ projects and 17M+ devices across multiple platforms
 - **OpenTelemetry pioneer** - Early advocate driving industry standards adoption in observability
 - **Multi-cloud infrastructure expertise** - Designed systems spanning AWS, GCP, and Azure at global scale
 - **Security contributor** - Contributed to discovery of CVE-2024-45337 in widely-used golang library
 - **24 years of progressive technical leadership** - From system administrator to Staff-level engineer
 - **Active open source contributor** - Early member of Python Argentina, conference speaker

Education

Universidad de Buenos Aires

1998–2007

Licenciado en Ciencias de la Computación

Thesis: Reinforcement Learning on Self-reconfigurable Modular Robots

Research Foundations

- On the possibilities of using previous knowledge in the form of Behaviour Trees in order to speed up Q-learning (2008)
- Thesis: Reinforcement Learning on Self-Reconfigurable Modular Robots (2007)
- State of the Art in the field of Self-reconfigurable Modular Robots (2005)

Leadership & Community

Python Argentina User Group

2005–Present

Early Member & Active Contributor

- **Co-organized PyCon Argentina** and PyCamp events
- **Speaker:** PyCon Argentina (2009, 2011, 2015, 2019), **PyCon US** (2012)
- Active contributor to Python community growth in Latin America

Languages: Spanish (Native), English (Fluent), German (Fluent)